

4 logarytmy 2

Oblicz:

a) $\log_{12} 2 + \log_{12} 8 + \log_{12} 9$

b) $\log 0,12 - \log 0,3 + \log 25$

a) $\log_{12} 2 + \log_{12} 8 + \log_{12} 9 = \log_{12} (2 \cdot 8 \cdot 9) =$

$$\log_a(x \cdot y) = \log_a x + \log_a y$$

$$= \log_{12} 144 = 2$$

Odp: 2.

b) $\log 0,12 - \log 0,3 + \log 25 =$

$$\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$$

$$= \log\left(\frac{0,12}{0,3}\right) + \log 25 = \log(0,4) + \log 25 =$$

$$\log_a(x \cdot y) = \log_a x + \log_a y$$

$$= \log(0,4 \cdot 25) = \log 10 = \log_{10} 10 = 1$$

logarytm bez
podstawy to
logarytm o
podstawie 10

Odp: 1.